

Fabric QA Code Challenge round





Overview:

- OrangeHRM is an open-source human resource management application that allows organizations to manage employee information, roles, and HR workflows.
- Develop a robust, end-to-end test automation framework using Playwright with TypeScript or JavaScript, covering both UI and API test scenarios along with CI/CD integration.
- Application URL: <https://opensource-demo.orangehrmlive.com/>

UI Test scenarios:

1. Navigate to the OrangeHRM application.
2. Login to the application using provided admin credentials.
3. Verify that the left-side navigation menu is displayed and functional.
4. Navigate to PIM (Personnel Information Management) module.
5. Add a new employee with valid details and capture the Employee ID.
6. Validate that the employee appears in the employee list with correct details.
7. Edit the employee details (e.g., job title, personal info) and validate updates.
8. Logout from the application.
9. Include assertions wherever applicable.

API Test scenarios:

1. Fetch the employee list using OrangeHRM API call.
2. Search for the employee created in UI using Employee ID.
3. Validate that UI data matches API response (name, job title, employee ID).
4. Include proper assertions to validate API response structure and content.



CICD Integration:

1. Integrate the Playwright test automation framework with GitHub Actions.
2. Configure the pipeline to:
 - Trigger automatically on every commit to the repository.
 - Run the full UI + API test suite.
3. After execution send a test report via email including:
 - Total tests executed
 - Number of passed and failed tests
 - Links to screenshots or logs for failed tests
 - Summary of retries or flaky tests

Note: Kindly consider below points while developing the test automation framework.

- Naming conventions, modularization, reusability and readability of code with appropriate project structure.
- Share us the GitHub Project link and reach out to us if any clarifications required.